SG No. 211 O^5 $I432$ [cubic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `2a`, site symmetry: `432`

Table 1: Wyckoff bond: `6a@2a`, bond symmetry: `42.2`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	[0, 0, 0]	`[1,-2,-3,4,17,-18,-19,20]`
2	$[0,\ X,\ 0]$	[0, 0, 0]	`[5,-6,-7,8,13,-14,-15,16]`
3	$[0,\ 0,\ X]$	[0, 0, 0]	`[9,-10,-11,12,-21,22,23,-24]`

Table 2: Wyckoff bond: `8b@2a`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X,\ X,\ X]$	[0, 0, 0]	`[1,5,9,-14,-19,-24]`
2	$[-X,\ -X,\ X]$	[0, 0, 0]	`[2,7,12,-13,-17,-21]`
3	$[-X,\ X,\ -X]$	[0, 0, 0]	`[3,8,10,-15,-20,-22]`
4	$[X,\ -X,\ -X]$	[0, 0, 0]	`[4,6,11,-16,-18,-23]`

Table 3: Wyckoff bond: `12c@2a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X,\ X,\ 0]$	[0, 0, 0]	`[1,-2,13,-14]`
2	$[-X,\ X,\ 0]$	[0, 0, 0]	`[3,-4,-15,16]`
3	$[0,\ X,\ X]$	[0, 0, 0]	`[5,-6,23,-24]`
4	$[0,\ -X,\ X]$	[0, 0, 0]	`[7,-8,-21,22]`
5	$[X,\ 0,\ X]$	[0, 0, 0]	`[9,-10,-19,20]`
6	$[X,\ 0,\ -X]$	[0, 0, 0]	`[11,-12,17,-18]`

Table 4: Wyckoff bond: `24d@2a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Y]$	[0, 0, 0]	`[1,-14]`
2	$[-X,\ -X,\ Y]$	[0, 0, 0]	`[2,-13]`
3	$[-X,\ X,\ -Y]$	[0, 0, 0]	`[3,-15]`
4	$[X,\ -X,\ -Y]$	[0, 0, 0]	`[4,-16]`
5	$[Y,\ X,\ X]$	[0, 0, 0]	`[5,-24]`
6	$[Y,\ -X,\ -X]$	[0, 0, 0]	`[6,-23]`
7	$[-Y,\ -X,\ X]$	[0, 0, 0]	`[7,-21]`
8	$[-Y,\ X,\ -X]$	[0, 0, 0]	`[8,-22]`
9	$[X,\ Y,\ X]$	[0, 0, 0]	`[9,-19]`
10	$[-X,\ Y,\ -X]$	[0, 0, 0]	`[10,-20]`
11	$[X,\ -Y,\ -X]$	[0, 0, 0]	`[11,-18]`
12	$[-X,\ -Y,\ X]$	[0, 0, 0]	`[12,-17]`

Table 5: Wyckoff bond: `24e@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	[0, 0, 0]	`[1,-2]`
2	$[-X, Y, 0]$	[0, 0, 0]	`[3,-4]`
3	$[0, X, Y]$	[0, 0, 0]	`[5,-6]`
4	$[0, -X, Y]$	[0, 0, 0]	`[7,-8]`
5	$[Y, 0, X]$	[0, 0, 0]	`[9,-10]`
6	$[Y, 0, -X]$	[0, 0, 0]	`[11,-12]`
7	$[Y, X, 0]$	[0, 0, 0]	`[13,-14]`
8	$[Y, -X, 0]$	[0, 0, 0]	`[15,-16]`
9	$[X, 0, -Y]$	[0, 0, 0]	`[17,-18]`
10	$[-X, 0, -Y]$	[0, 0, 0]	`[19,-20]`
11	$[0, Y, -X]$	[0, 0, 0]	`[21,-22]`
12	$[0, Y, X]$	[0, 0, 0]	`[23,-24]`

Table 6: Wyckoff bond: `48f@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	[0, 0, 0]	`[1]`
2	$[-X, -Y, Z]$	[0, 0, 0]	`[2]`
3	$[-X, Y, -Z]$	[0, 0, 0]	`[3]`
4	$[X, -Y, -Z]$	[0, 0, 0]	`[4]`
5	$[Z, X, Y]$	[0, 0, 0]	`[5]`
6	$[Z, -X, -Y]$	[0, 0, 0]	`[6]`
7	$[-Z, -X, Y]$	[0, 0, 0]	`[7]`
8	$[-Z, X, -Y]$	[0, 0, 0]	`[8]`
9	$[Y, Z, X]$	[0, 0, 0]	`[9]`
10	$[-Y, Z, -X]$	[0, 0, 0]	`[10]`
11	$[Y, -Z, -X]$	[0, 0, 0]	`[11]`
12	$[-Y, -Z, X]$	[0, 0, 0]	`[12]`
13	$[Y, X, -Z]$	[0, 0, 0]	`[13]`
14	$[-Y, -X, -Z]$	[0, 0, 0]	`[14]`
15	$[Y, -X, Z]$	[0, 0, 0]	`[15]`
16	$[-Y, X, Z]$	[0, 0, 0]	`[16]`
17	$[X, Z, -Y]$	[0, 0, 0]	`[17]`
18	$[-X, Z, Y]$	[0, 0, 0]	`[18]`
19	$[-X, -Z, -Y]$	[0, 0, 0]	`[19]`
20	$[X, -Z, Y]$	[0, 0, 0]	`[20]`
21	$[Z, Y, -X]$	[0, 0, 0]	`[21]`
22	$[Z, -Y, X]$	[0, 0, 0]	`[22]`
23	$[-Z, Y, X]$	[0, 0, 0]	`[23]`
24	$[-Z, -Y, -X]$	[0, 0, 0]	`[24]`

* Wyckoff site: `6b`, site symmetry: `42.2`

Table 7: Wyckoff bond: `6a@6b`, bond symmetry: `42.2`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2,-3,4,17,-18,-19,20]`
2	$[0, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6,-7,8,13,-14,-15,16]`
3	$[0, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10,-11,12,-21,22,23,-24]`

Table 8: Wyckoff bond: `12b@6b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4,18,-19]`
2	$[0, -X, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3,-17,20]`
3	$[X, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-8,-14,15]`
4	$[X, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-7,13,-16]`
5	$[X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-12,21,-24]`
6	$[-X, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-11,-22,23]`

Table 9: Wyckoff bond: `12c@6b`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[0, 0, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,-3,-4]`
2	$[X, 0, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,-7,-8]`
3	$[0, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,-11,-12]`
4	$[0, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,14,-15,-16]`
5	$[0, X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[17,18,-19,-20]`
6	$[X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[21,22,-23,-24]`

Table 10: Wyckoff bond: `24d@6b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-19]`
2	$[-Y, -X, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-17]`
3	$[-Y, X, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-20]`
4	$[Y, -X, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-18]`
5	$[X, Y, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-14]`
6	$[X, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-16]`
7	$[-X, -Y, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-13]`
8	$[-X, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8,-15]`
9	$[X, X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-24]`
10	$[-X, X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-22]`
11	$[X, -X, -Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,-23]`
12	$[-X, -X, Y]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12,-21]`

Table 11: Wyckoff bond: `24e@6b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,-4]`
3	$[0, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-6]`
4	$[0, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,-8]`
5	$[Y, 0, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-10]`
6	$[Y, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11,-12]`
7	$[Y, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,-14]`
8	$[Y, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[15,-16]`
9	$[X, 0, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-18]`
10	$[-X, 0, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[19,-20]`
11	$[0, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[21,-22]`
12	$[0, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[23,-24]`

Table 12: Wyckoff bond: `24f@6b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[22,-23]`

Table 13: Wyckoff bond: `48g@6b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[8]`
9	$[Y, Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[12]`

continued ...

Table 13

No.	vector	center	mapping
13	$[Y, X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[16]
17	$[X, Z, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[17]
18	$[-X, Z, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[19]
20	$[X, -Z, Y]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[24]

* Wyckoff site: 8c, site symmetry: .32

Table 14: Wyckoff bond: 8a@8c, bond symmetry: .32

No.	vector	center	mapping
1	$[X, X, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,5,9,-14,-19,-24]
2	$[-X, -X, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,7,12,-13,-17,-21]
3	$[-X, X, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,8,10,-15,-20,-22]
4	$[X, -X, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,6,11,-16,-18,-23]

Table 15: Wyckoff bond: 24b@8c, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,-14]
2	$[-X, -X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,-13]
3	$[-X, X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,-15]
4	$[X, -X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,-16]
5	$[Y, X, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5,-24]
6	$[Y, -X, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[6,-23]
7	$[-Y, -X, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[7,-21]
8	$[-Y, X, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8,-22]
9	$[X, Y, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[9,-19]
10	$[-X, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[10,-20]
11	$[X, -Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[11,-18]
12	$[-X, -Y, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[12,-17]

Table 16: Wyckoff bond: 24c@8c, bond symmetry: ..2

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,14]
2	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,13]

continued ...

Table 16

No.	vector	center	mapping
3	$[-X, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3,15]
4	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,16]
5	$[0, X, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5,24]
6	$[0, -X, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[6,23]
7	$[0, -X, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[7,21]
8	$[0, X, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8,22]
9	$[-X, 0, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[9,19]
10	$[X, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[10,20]
11	$[-X, 0, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[11,18]
12	$[X, 0, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[12,17]

Table 17: Wyckoff bond: 48d@8c, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8]
9	$[Y, Z, X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[24]

* Wyckoff site: 12d, site symmetry: 2.22

Table 18: Wyckoff bond: 12a@12d, bond symmetry: 2.22

No.	vector	center	mapping
1	$[0, X, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	[1,-4,18,-19]
2	$[0, -X, X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	[2,-3,-17,20]

continued ...

Table 18

No.	vector	center	mapping
3	$[X, 0, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,-8,-14,15]`
4	$[X, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,-7,13,-16]`
5	$[X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,-12,21,-24]`
6	$[-X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,-11,-22,23]`

Table 19: Wyckoff bond: `12b@12d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,-4,-18,19]`
2	$[0, -X, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,-3,17,-20]`
3	$[-X, 0, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,-8,14,-15]`
4	$[-X, 0, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,-7,-13,16]`
5	$[X, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,-12,-21,24]`
6	$[-X, -X, 0]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,-11,22,-23]`

Table 20: Wyckoff bond: `12c@12d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,4,-18,-19]`
2	$[-X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,3,-17,-20]`
3	$[0, X, 0]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,8,-14,-15]`
4	$[0, -X, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,-13,-16]`
5	$[0, 0, X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,12,-21,-24]`
6	$[0, 0, -X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,-22,-23]`

Table 21: Wyckoff bond: `24d@12d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,-19]`
2	$[-Y, -X, X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,-17]`
3	$[-Y, X, -X]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[3,-20]`
4	$[Y, -X, -X]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[4,-18]`
5	$[X, Y, X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,-14]`
6	$[X, -Y, -X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,-16]`
7	$[-X, -Y, X]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[7,-13]`
8	$[-X, Y, -X]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[8,-15]`
9	$[X, X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,-24]`
10	$[-X, X, -Y]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,-22]`
11	$[X, -X, -Y]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[11,-23]`
12	$[-X, -X, Y]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[12,-21]`

Table 22: Wyckoff bond: `24e@12d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[Y, X, -X]$	$[\frac{1}{4}, \frac{1}{2}, 0]$	`[1,-18]`
2	$[-Y, -X, -X]$	$[\frac{3}{4}, \frac{1}{2}, 0]$	`[2,-20]`
3	$[-Y, X, X]$	$[\frac{3}{4}, \frac{1}{2}, 0]$	`[3,-17]`
4	$[Y, -X, X]$	$[\frac{1}{4}, \frac{1}{2}, 0]$	`[4,-19]`
5	$[-X, Y, X]$	$[0, \frac{1}{4}, \frac{1}{2}]$	`[5,-15]`
6	$[-X, -Y, -X]$	$[0, \frac{3}{4}, \frac{1}{2}]$	`[6,-13]`
7	$[X, -Y, X]$	$[0, \frac{3}{4}, \frac{1}{2}]$	`[7,-16]`
8	$[X, Y, -X]$	$[0, \frac{1}{4}, \frac{1}{2}]$	`[8,-14]`
9	$[X, -X, Y]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[9,-21]`
10	$[-X, -X, -Y]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[10,-23]`
11	$[X, X, -Y]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[11,-22]`
12	$[-X, X, Y]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[12,-24]`

Table 23: Wyckoff bond: `24f@12d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[\frac{1}{4}, \frac{1}{2}, 0]$	`[1,-4]`
2	$[0, -X, Y]$	$[\frac{3}{4}, \frac{1}{2}, 0]$	`[2,-3]`
3	$[Y, 0, X]$	$[0, \frac{1}{4}, \frac{1}{2}]$	`[5,-8]`
4	$[Y, 0, -X]$	$[0, \frac{3}{4}, \frac{1}{2}]$	`[6,-7]`
5	$[X, Y, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[9,-12]`
6	$[-X, Y, 0]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[10,-11]`
7	$[X, 0, -Y]$	$[0, \frac{3}{4}, \frac{1}{2}]$	`[13,-16]`
8	$[-X, 0, -Y]$	$[0, \frac{1}{4}, \frac{1}{2}]$	`[14,-15]`
9	$[0, Y, -X]$	$[\frac{3}{4}, \frac{1}{2}, 0]$	`[17,-20]`
10	$[0, Y, X]$	$[\frac{1}{4}, \frac{1}{2}, 0]$	`[18,-19]`
11	$[Y, X, 0]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[21,-24]`
12	$[Y, -X, 0]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[22,-23]`

Table 24: Wyckoff bond: `48g@12d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{1}{2}, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{2}, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{2}, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{1}{2}, 0]$	`[4]`
5	$[Z, X, Y]$	$[0, \frac{1}{4}, \frac{1}{2}]$	`[5]`
6	$[Z, -X, -Y]$	$[0, \frac{3}{4}, \frac{1}{2}]$	`[6]`
7	$[-Z, -X, Y]$	$[0, \frac{3}{4}, \frac{1}{2}]$	`[7]`
8	$[-Z, X, -Y]$	$[0, \frac{1}{4}, \frac{1}{2}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{1}{2}, 0, \frac{3}{4}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{1}{2}, 0, \frac{1}{4}]$	`[12]`

continued ...

Table 24

No.	vector	center	mapping
13	$[Y, X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[24]`

* Wyckoff site: `12e`, site symmetry: `4..`

Table 25: Wyckoff bond: `12a@12e`, bond symmetry: `4..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4,17,20]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,3,18,19]`
3	$[0, X, 0]$	$[0, x, 0]$	`[5,8,13,16]`
4	$[0, -X, 0]$	$[0, -x, 0]$	`[6,7,14,15]`
5	$[0, 0, X]$	$[0, 0, x]$	`[9,12,22,23]`
6	$[0, 0, -X]$	$[0, 0, -x]$	`[10,11,21,24]`

Table 26: Wyckoff bond: `24b@12e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -X, Y]$	$[-x, 0, 0]$	`[2,-3]`
3	$[Y, 0, X]$	$[0, x, 0]$	`[5,-8]`
4	$[Y, 0, -X]$	$[0, -x, 0]$	`[6,-7]`
5	$[X, Y, 0]$	$[0, 0, x]$	`[9,-12]`
6	$[-X, Y, 0]$	$[0, 0, -x]$	`[10,-11]`
7	$[X, 0, -Y]$	$[0, x, 0]$	`[13,-16]`
8	$[-X, 0, -Y]$	$[0, -x, 0]$	`[14,-15]`
9	$[0, Y, -X]$	$[x, 0, 0]$	`[17,-20]`
10	$[0, Y, X]$	$[-x, 0, 0]$	`[18,-19]`
11	$[Y, X, 0]$	$[0, 0, -x]$	`[21,-24]`
12	$[Y, -X, 0]$	$[0, 0, x]$	`[22,-23]`

Table 27: Wyckoff bond: `48c@12e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[Z, X, Y]$	$[0, x, 0]$	`[5]`
6	$[Z, -X, -Y]$	$[0, -x, 0]$	`[6]`
7	$[-Z, -X, Y]$	$[0, -x, 0]$	`[7]`
8	$[-Z, X, -Y]$	$[0, x, 0]$	`[8]`
9	$[Y, Z, X]$	$[0, 0, x]$	`[9]`
10	$[-Y, Z, -X]$	$[0, 0, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[0, 0, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[0, 0, x]$	`[12]`
13	$[Y, X, -Z]$	$[0, x, 0]$	`[13]`
14	$[-Y, -X, -Z]$	$[0, -x, 0]$	`[14]`
15	$[Y, -X, Z]$	$[0, -x, 0]$	`[15]`
16	$[-Y, X, Z]$	$[0, x, 0]$	`[16]`
17	$[X, Z, -Y]$	$[x, 0, 0]$	`[17]`
18	$[-X, Z, Y]$	$[-x, 0, 0]$	`[18]`
19	$[-X, -Z, -Y]$	$[-x, 0, 0]$	`[19]`
20	$[X, -Z, Y]$	$[x, 0, 0]$	`[20]`
21	$[Z, Y, -X]$	$[0, 0, -x]$	`[21]`
22	$[Z, -Y, X]$	$[0, 0, x]$	`[22]`
23	$[-Z, Y, X]$	$[0, 0, x]$	`[23]`
24	$[-Z, -Y, -X]$	$[0, 0, -x]$	`[24]`

* Wyckoff site: `16f`, site symmetry: `.3.`

Table 28: Wyckoff bond: `16a@16f`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	`[1,5,9]`
2	$[-X, -X, X]$	$[-x, -x, x]$	`[2,7,12]`
3	$[-X, X, -X]$	$[-x, x, -x]$	`[3,8,10]`
4	$[X, -X, -X]$	$[x, -x, -x]$	`[4,6,11]`
5	$[X, X, -X]$	$[x, x, -x]$	`[13,17,21]`
6	$[-X, -X, -X]$	$[-x, -x, -x]$	`[14,19,24]`
7	$[X, -X, X]$	$[x, -x, x]$	`[15,20,22]`
8	$[-X, X, X]$	$[-x, x, x]$	`[16,18,23]`

Table 29: Wyckoff bond: `48b@16f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, x]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, x, -x]$	`[3]`

continued ...

Table 29

No.	vector	center	mapping
4	$[X, -Y, -Z]$	$[x, -x, -x]$	`[4]`
5	$[Z, X, Y]$	$[x, x, x]$	`[5]`
6	$[Z, -X, -Y]$	$[x, -x, -x]$	`[6]`
7	$[-Z, -X, Y]$	$[-x, -x, x]$	`[7]`
8	$[-Z, X, -Y]$	$[-x, x, -x]$	`[8]`
9	$[Y, Z, X]$	$[x, x, x]$	`[9]`
10	$[-Y, Z, -X]$	$[-x, x, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[x, -x, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[-x, -x, x]$	`[12]`
13	$[Y, X, -Z]$	$[x, x, -x]$	`[13]`
14	$[-Y, -X, -Z]$	$[-x, -x, -x]$	`[14]`
15	$[Y, -X, Z]$	$[x, -x, x]$	`[15]`
16	$[-Y, X, Z]$	$[-x, x, x]$	`[16]`
17	$[X, Z, -Y]$	$[x, x, -x]$	`[17]`
18	$[-X, Z, Y]$	$[-x, x, x]$	`[18]`
19	$[-X, -Z, -Y]$	$[-x, -x, -x]$	`[19]`
20	$[X, -Z, Y]$	$[x, -x, x]$	`[20]`
21	$[Z, Y, -X]$	$[x, x, -x]$	`[21]`
22	$[Z, -Y, X]$	$[x, -x, x]$	`[22]`
23	$[-Z, Y, X]$	$[-x, x, x]$	`[23]`
24	$[-Z, -Y, -X]$	$[-x, -x, -x]$	`[24]`

* Wyckoff site: `24g`, site symmetry: `2..`

Table 30: Wyckoff bond: `24a@24g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,-3]`
3	$[Y, 0, X]$	$\left[0, x, \frac{1}{2}\right]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[0, -x, \frac{1}{2}\right]$	`[6,-7]`
5	$[X, Y, 0]$	$\left[\frac{1}{2}, 0, x\right]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, 0, -x\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{1}{2}, x, 0\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{1}{2}, -x, 0\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[x, 0, \frac{1}{2}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[-x, 0, \frac{1}{2}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[0, \frac{1}{2}, -x\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[0, \frac{1}{2}, x\right]$	`[22,-23]`

Table 31: Wyckoff bond: `24b@24g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[x, \frac{1}{2}, 0\right]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, 0\right]$	`[2,3]`

continued ...

Table 31

No.	vector	center	mapping
3	$[0,\ X,\ 0]$	$\left[0,\ x,\ \frac{1}{2}\right]$	[5,8]
4	$[0,\ -X,\ 0]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	[6,7]
5	$[0,\ 0,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	[9,12]
6	$[0,\ 0,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	[10,11]
7	$[0,\ X,\ 0]$	$\left[\frac{1}{2},\ x,\ 0\right]$	[13,16]
8	$[0,\ -X,\ 0]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	[14,15]
9	$[X,\ 0,\ 0]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	[17,20]
10	$[-X,\ 0,\ 0]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[18,19]
11	$[0,\ 0,\ -X]$	$\left[0,\ \frac{1}{2},\ -x\right]$	[21,24]
12	$[0,\ 0,\ X]$	$\left[0,\ \frac{1}{2},\ x\right]$	[22,23]

Table 32: Wyckoff bond: 48c@24g, bond symmetry: 1

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	[1]
2	$[-X,\ -Y,\ Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	[2]
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ \frac{1}{2},\ 0\right]$	[3]
4	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{2},\ 0\right]$	[4]
5	$[Z,\ X,\ Y]$	$\left[0,\ x,\ \frac{1}{2}\right]$	[5]
6	$[Z,\ -X,\ -Y]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	[6]
7	$[-Z,\ -X,\ Y]$	$\left[0,\ -x,\ \frac{1}{2}\right]$	[7]
8	$[-Z,\ X,\ -Y]$	$\left[0,\ x,\ \frac{1}{2}\right]$	[8]
9	$[Y,\ Z,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	[9]
10	$[-Y,\ Z,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	[10]
11	$[Y,\ -Z,\ -X]$	$\left[\frac{1}{2},\ 0,\ -x\right]$	[11]
12	$[-Y,\ -Z,\ X]$	$\left[\frac{1}{2},\ 0,\ x\right]$	[12]
13	$[Y,\ X,\ -Z]$	$\left[\frac{1}{2},\ x,\ 0\right]$	[13]
14	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	[14]
15	$[Y,\ -X,\ Z]$	$\left[\frac{1}{2},\ -x,\ 0\right]$	[15]
16	$[-Y,\ X,\ Z]$	$\left[\frac{1}{2},\ x,\ 0\right]$	[16]
17	$[X,\ Z,\ -Y]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	[17]
18	$[-X,\ Z,\ Y]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[18]
19	$[-X,\ -Z,\ -Y]$	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[19]
20	$[X,\ -Z,\ Y]$	$\left[x,\ 0,\ \frac{1}{2}\right]$	[20]
21	$[Z,\ Y,\ -X]$	$\left[0,\ \frac{1}{2},\ -x\right]$	[21]
22	$[Z,\ -Y,\ X]$	$\left[0,\ \frac{1}{2},\ x\right]$	[22]
23	$[-Z,\ Y,\ X]$	$\left[0,\ \frac{1}{2},\ x\right]$	[23]
24	$[-Z,\ -Y,\ -X]$	$\left[0,\ \frac{1}{2},\ -x\right]$	[24]

* Wyckoff site: 24h, site symmetry: ..2

Table 33: Wyckoff bond: 24a@24h, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y,\ X,\ -X]$	$[0,\ y,\ y]$	[1,-18]
2	$[-Y,\ -X,\ -X]$	$[0,\ -y,\ y]$	[2,-20]

continued ...

Table 33

No.	vector	center	mapping
3	$[-Y, X, X]$	$[0, y, -y]$	`[3,-17]`
4	$[Y, -X, X]$	$[0, -y, -y]$	`[4,-19]`
5	$[-X, Y, X]$	$[y, 0, y]$	`[5,-15]`
6	$[-X, -Y, -X]$	$[y, 0, -y]$	`[6,-13]`
7	$[X, -Y, X]$	$[-y, 0, y]$	`[7,-16]`
8	$[X, Y, -X]$	$[-y, 0, -y]$	`[8,-14]`
9	$[X, -X, Y]$	$[y, y, 0]$	`[9,-21]`
10	$[-X, -X, -Y]$	$[-y, y, 0]$	`[10,-23]`
11	$[X, X, -Y]$	$[y, -y, 0]$	`[11,-22]`
12	$[-X, X, Y]$	$[-y, -y, 0]$	`[12,-24]`

Table 34: Wyckoff bond: `24b@24h`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0, X, X]$	$[0, y, y]$	`[1,18]`
2	$[0, -X, X]$	$[0, -y, y]$	`[2,20]`
3	$[0, X, -X]$	$[0, y, -y]$	`[3,17]`
4	$[0, -X, -X]$	$[0, -y, -y]$	`[4,19]`
5	$[X, 0, X]$	$[y, 0, y]$	`[5,15]`
6	$[X, 0, -X]$	$[y, 0, -y]$	`[6,13]`
7	$[-X, 0, X]$	$[-y, 0, y]$	`[7,16]`
8	$[-X, 0, -X]$	$[-y, 0, -y]$	`[8,14]`
9	$[X, X, 0]$	$[y, y, 0]$	`[9,21]`
10	$[-X, X, 0]$	$[-y, y, 0]$	`[10,23]`
11	$[X, -X, 0]$	$[y, -y, 0]$	`[11,22]`
12	$[-X, -X, 0]$	$[-y, -y, 0]$	`[12,24]`

Table 35: Wyckoff bond: `48c@24h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, y]$	`[1]`
2	$[-X, -Y, Z]$	$[0, -y, y]$	`[2]`
3	$[-X, Y, -Z]$	$[0, y, -y]$	`[3]`
4	$[X, -Y, -Z]$	$[0, -y, -y]$	`[4]`
5	$[Z, X, Y]$	$[y, 0, y]$	`[5]`
6	$[Z, -X, -Y]$	$[y, 0, -y]$	`[6]`
7	$[-Z, -X, Y]$	$[-y, 0, y]$	`[7]`
8	$[-Z, X, -Y]$	$[-y, 0, -y]$	`[8]`
9	$[Y, Z, X]$	$[y, y, 0]$	`[9]`
10	$[-Y, Z, -X]$	$[-y, y, 0]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -y, 0]$	`[11]`
12	$[-Y, -Z, X]$	$[-y, -y, 0]$	`[12]`
13	$[Y, X, -Z]$	$[y, 0, -y]$	`[13]`
14	$[-Y, -X, -Z]$	$[-y, 0, -y]$	`[14]`

continued ...

Table 35

No.	vector	center	mapping
15	$[Y, -X, Z]$	$[y, 0, y]$	[15]
16	$[-Y, X, Z]$	$[-y, 0, y]$	[16]
17	$[X, Z, -Y]$	$[0, y, -y]$	[17]
18	$[-X, Z, Y]$	$[0, y, y]$	[18]
19	$[-X, -Z, -Y]$	$[0, -y, -y]$	[19]
20	$[X, -Z, Y]$	$[0, -y, y]$	[20]
21	$[Z, Y, -X]$	$[y, y, 0]$	[21]
22	$[Z, -Y, X]$	$[y, -y, 0]$	[22]
23	$[-Z, Y, X]$	$[-y, y, 0]$	[23]
24	$[-Z, -Y, -X]$	$[-y, -y, 0]$	[24]

* Wyckoff site: 24i, site symmetry: ..2

Table 36: Wyckoff bond: 24a@24i, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{4}, y, \frac{1}{2} - y\right]$	[1,-19]
2	$[-Y, -X, X]$	$\left[\frac{3}{4}, -y, \frac{1}{2} - y\right]$	[2,-17]
3	$[-Y, X, -X]$	$\left[\frac{3}{4}, y, y + \frac{1}{2}\right]$	[3,-20]
4	$[Y, -X, -X]$	$\left[\frac{1}{4}, -y, y + \frac{1}{2}\right]$	[4,-18]
5	$[X, Y, X]$	$\left[\frac{1}{2} - y, \frac{1}{4}, y\right]$	[5,-14]
6	$[X, -Y, -X]$	$\left[\frac{1}{2} - y, \frac{3}{4}, -y\right]$	[6,-16]
7	$[-X, -Y, X]$	$\left[y + \frac{1}{2}, \frac{3}{4}, y\right]$	[7,-13]
8	$[-X, Y, -X]$	$\left[y + \frac{1}{2}, \frac{1}{4}, -y\right]$	[8,-15]
9	$[X, X, Y]$	$\left[y, \frac{1}{2} - y, \frac{1}{4}\right]$	[9,-24]
10	$[-X, X, -Y]$	$\left[-y, \frac{1}{2} - y, \frac{3}{4}\right]$	[10,-22]
11	$[X, -X, -Y]$	$\left[y, y + \frac{1}{2}, \frac{3}{4}\right]$	[11,-23]
12	$[-X, -X, Y]$	$\left[-y, y + \frac{1}{2}, \frac{1}{4}\right]$	[12,-21]

Table 37: Wyckoff bond: 24b@24i, bond symmetry: ..2

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{1}{4}, y, \frac{1}{2} - y\right]$	[1,19]
2	$[0, -X, -X]$	$\left[\frac{3}{4}, -y, \frac{1}{2} - y\right]$	[2,17]
3	$[0, X, X]$	$\left[\frac{3}{4}, y, y + \frac{1}{2}\right]$	[3,20]
4	$[0, -X, X]$	$\left[\frac{1}{4}, -y, y + \frac{1}{2}\right]$	[4,18]
5	$[-X, 0, X]$	$\left[\frac{1}{2} - y, \frac{1}{4}, y\right]$	[5,14]
6	$[-X, 0, -X]$	$\left[\frac{1}{2} - y, \frac{3}{4}, -y\right]$	[6,16]
7	$[X, 0, X]$	$\left[y + \frac{1}{2}, \frac{3}{4}, y\right]$	[7,13]
8	$[X, 0, -X]$	$\left[y + \frac{1}{2}, \frac{1}{4}, -y\right]$	[8,15]
9	$[X, -X, 0]$	$\left[y, \frac{1}{2} - y, \frac{1}{4}\right]$	[9,24]
10	$[-X, -X, 0]$	$\left[-y, \frac{1}{2} - y, \frac{3}{4}\right]$	[10,22]
11	$[X, X, 0]$	$\left[y, y + \frac{1}{2}, \frac{3}{4}\right]$	[11,23]
12	$[-X, X, 0]$	$\left[-y, y + \frac{1}{2}, \frac{1}{4}\right]$	[12,21]

Table 38: Wyckoff bond: `48c@24i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, -y, \frac{1}{2}-y\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{3}{4}, y, y+\frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, -y, y+\frac{1}{2}\right]$	`[4]`
5	$[Z, X, Y]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	`[5]`
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}-y, \frac{3}{4}, -y\right]$	`[6]`
7	$[-Z, -X, Y]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y\right]$	`[7]`
8	$[-Z, X, -Y]$	$\left[y+\frac{1}{2}, \frac{1}{4}, -y\right]$	`[8]`
9	$[Y, Z, X]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	`[9]`
10	$[-Y, Z, -X]$	$\left[-y, \frac{1}{2}-y, \frac{3}{4}\right]$	`[10]`
11	$[Y, -Z, -X]$	$\left[y, y+\frac{1}{2}, \frac{3}{4}\right]$	`[11]`
12	$[-Y, -Z, X]$	$\left[-y, y+\frac{1}{2}, \frac{1}{4}\right]$	`[12]`
13	$[Y, X, -Z]$	$\left[y+\frac{1}{2}, \frac{3}{4}, y\right]$	`[13]`
14	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-y, \frac{1}{4}, y\right]$	`[14]`
15	$[Y, -X, Z]$	$\left[y+\frac{1}{2}, \frac{1}{4}, -y\right]$	`[15]`
16	$[-Y, X, Z]$	$\left[\frac{1}{2}-y, \frac{3}{4}, -y\right]$	`[16]`
17	$[X, Z, -Y]$	$\left[\frac{3}{4}, -y, \frac{1}{2}-y\right]$	`[17]`
18	$[-X, Z, Y]$	$\left[\frac{1}{4}, -y, y+\frac{1}{2}\right]$	`[18]`
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4}, y, \frac{1}{2}-y\right]$	`[19]`
20	$[X, -Z, Y]$	$\left[\frac{3}{4}, y, y+\frac{1}{2}\right]$	`[20]`
21	$[Z, Y, -X]$	$\left[-y, y+\frac{1}{2}, \frac{1}{4}\right]$	`[21]`
22	$[Z, -Y, X]$	$\left[-y, \frac{1}{2}-y, \frac{3}{4}\right]$	`[22]`
23	$[-Z, Y, X]$	$\left[y, y+\frac{1}{2}, \frac{3}{4}\right]$	`[23]`
24	$[-Z, -Y, -X]$	$\left[y, \frac{1}{2}-y, \frac{1}{4}\right]$	`[24]`

* Wyckoff site: `48j`, site symmetry: `1`

Table 39: Wyckoff bond: `48a@48j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-X, Y, -Z]$	$[-x, y, -z]$	`[3]`
4	$[X, -Y, -Z]$	$[x, -y, -z]$	`[4]`
5	$[Z, X, Y]$	$[z, x, y]$	`[5]`
6	$[Z, -X, -Y]$	$[z, -x, -y]$	`[6]`
7	$[-Z, -X, Y]$	$[-z, -x, y]$	`[7]`
8	$[-Z, X, -Y]$	$[-z, x, -y]$	`[8]`
9	$[Y, Z, X]$	$[y, z, x]$	`[9]`
10	$[-Y, Z, -X]$	$[-y, z, -x]$	`[10]`
11	$[Y, -Z, -X]$	$[y, -z, -x]$	`[11]`
12	$[-Y, -Z, X]$	$[-y, -z, x]$	`[12]`
13	$[Y, X, -Z]$	$[y, x, -z]$	`[13]`
14	$[-Y, -X, -Z]$	$[-y, -x, -z]$	`[14]`
15	$[Y, -X, Z]$	$[y, -x, z]$	`[15]`
16	$[-Y, X, Z]$	$[-y, x, z]$	`[16]`

continued ...

Table 39

No.	vector	center	mapping
17	$[X, Z, -Y]$	$[x, z, -y]$	[17]
18	$[-X, Z, Y]$	$[-x, z, y]$	[18]
19	$[-X, -Z, -Y]$	$[-x, -z, -y]$	[19]
20	$[X, -Z, Y]$	$[x, -z, y]$	[20]
21	$[Z, Y, -X]$	$[z, y, -x]$	[21]
22	$[Z, -Y, X]$	$[z, -y, x]$	[22]
23	$[-Z, Y, X]$	$[-z, y, x]$	[23]
24	$[-Z, -Y, -X]$	$[-z, -y, -x]$	[24]